Set 22: Time Series Analysis

AIM

To analyze monthly sales data using R programming by creating a time series line chart, a scatter plot, and an autocorrelation plot to identify sales trends and seasonality.

ALGORITHM

1. Create a dataset containing Month and Sales.
2. Load the ggplot2 library.
3. Convert the monthly sales data into a time series object.
4. Create a line chart to visualize monthly sales trends.
5. Create a scatter plot to examine the relationship between advertising budget and monthly sales.
6. Use Tableau to combine the line chart, scatter plot, and pie chart into a dashboard.
7. Generate an autocorrelation plot to identify patterns in the sales data.
8. Display the charts and interpret the results.

PROGRAM

```
library(ggplot2)


sales <- data.frame(
  Month = c("January", "February", "March", "April", "May"),
  Sales = c(15000, 18000, 22000, 20000, 23000),
  Advertising = c(2000, 2500, 3000, 2800, 3500)
)


sales$Month <- factor(sales$Month, levels = sales$Month)

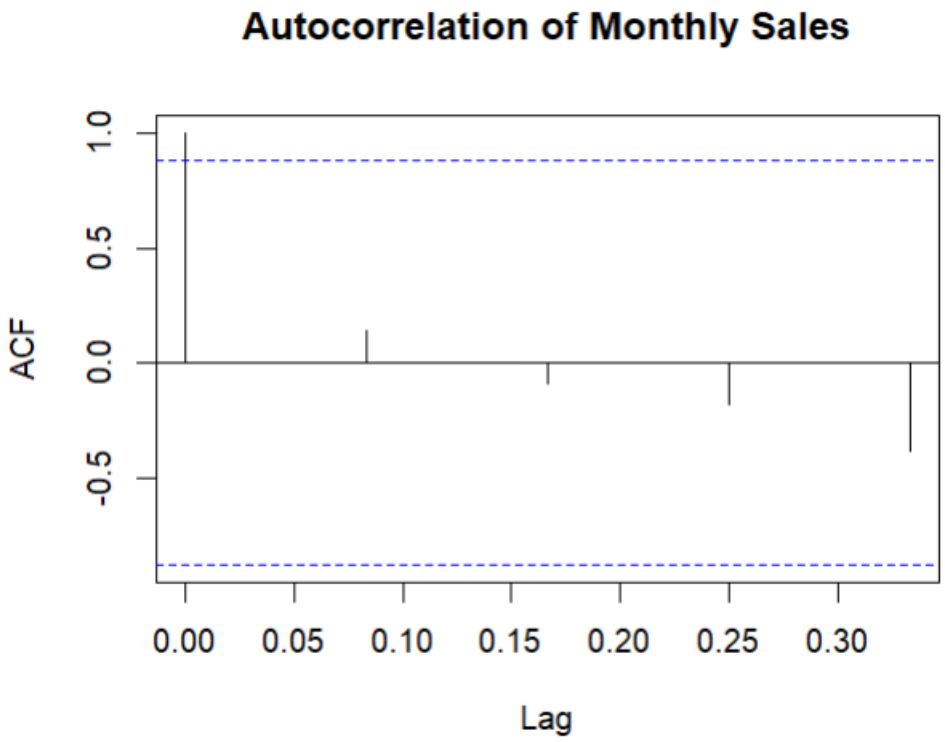

ggplot(sales, aes(Month, Sales, group = 1)) +
  geom_line(color = "blue") +
  geom_point() +
  labs(title = "Monthly Sales Trend",
       x = "Month", y = "Sales ($)")
```

```
ggplot(sales, aes(Advertising, Sales)) +
  geom_point(color = "red", size = 3) +
  labs(title = "Advertising Budget vs Monthly Sales",
       x = "Advertising Budget ($)", y = "Sales ($)")


sales_ts <- ts(sales$Sales, start = c(2023, 1), frequency = 12)
acf(sales_ts, main = "Autocorrelation of Monthly Sales")
```

Note: The advertising budget values are illustrative because they are not included in the given dataset. Replace them with actual values if provided.

OUTPUT:



RESULT

The line chart shows that sales generally increased from January to May, with a decrease in April. The scatter plot illustrates the relationship between advertising expenditure and sales. The autocorrelation plot helps examine relationships between sales at different time lags; however, five months of data are insufficient to establish reliable yearly seasonality.